

# SARVESH GANESAN

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## PROFESSIONAL SUMMARY

Lead AI Architect with 3+ years of experience building and shipping cloud-native AI and data platforms. Architected 12+ production microservices spanning agentic AI, distributed ETL, data lakehousing, and MLOps on AWS EKS. Proven track record of turning ambiguous product requirements into scalable, cost-optimized systems for enterprise workloads.

## TECHNICAL SKILLS

- **Languages & Frameworks:** Python, FastAPI, Flask, REST APIs, gRPC, SSE streaming
- **AI / LLM:** AWS Bedrock, Strands Agents, LangChain, LangGraph, RAG systems, MCP
- **Data Engineering:** Apache Spark, Apache Iceberg, DuckDB, Pandas, PyArrow, Polars
- **Databases:** PostgreSQL, pgvector, Redis, Pinecone, MongoDB, Elasticsearch, DynamoDB
- **Cloud & Infra:** AWS (EKS, Bedrock, S3, Lambda, ECR, CloudWatch), Docker, Kubernetes, Karpenter
- **MLOps:** MLflow, Jupyter, PyTorch, TensorFlow, scikit-learn, Docker-based GPU runtimes

## EXPERIENCE

### Lead AI Architect

Sep 2023 - Mar 2026

Groundzero Software Private Limited

Chennai, India

- Owned the entire backend architecture end-to-end, designing and shipping 12+ production microservices for agentic chat, data providers, ETL orchestration, Iceberg lakehouse APIs, MLOps, and Spark/DuckDB runtimes as the sole backend engineer.
- Built a multi-agent AI system that lets enterprise users query databases, build dashboards, run ETL jobs, and manage notebooks through natural language, reducing time-to-insight from hours of manual work to seconds of conversation.
- Scaled data connectivity to 69+ sources through a dynamically loaded connector architecture with role-based authorization and schema introspection caching.
- Reduced RAG query latency by 90% using hybrid vector plus full-text retrieval with neural reranking, enabling real-time document Q&A; at production scale.
- Delivered sub-10s query performance on 100M+ row datasets by architecting a TB-scale Apache Iceberg lakehouse with Spark and DuckDB compute engines.
- Cut LLM inference costs with three-tier model routing, Anthropic prompt caching, and per-request token tracking across agent workflows.
- Designed the ETL orchestration layer for job submission, multi-job chaining, container log streaming from EKS pods, and dataset lineage across Spark, DuckDB, and Pandas engines.
- Stood up MLOps and platform infrastructure with MLflow, Jupyter lifecycle management on EKS, GPU training containers, Karpenter autoscaling, and 99.9% uptime across multi-tenant deployments.

### Software Trainee

Jun 2023 - Aug 2023

SCI-BI Software Solutions Private Limited

Chennai, India

- Built client-facing dashboards and reports in Power BI and Tableau, and translated client requirements into technical specifications.

## PROJECTS

### PayPal Agent | Python, LangGraph, FastAPI, pgvector, AWS Bedrock

GitHub

- Built a PayPal tool router and execution service covering 116 API operations with multi-provider model routing, hybrid RAG, guarded mutation flows, OAuth token caching, and a terminal UI.
- Added strict Pydantic routing, sandbox execution, metadata-only memory, LangSmith tracing, and recorded demo artifacts for a complete, verifiable workflow.

### SRE Tea MCP Server | Node.js, Model Context Protocol, Claude, JWT, AWS Lambda

GitHub

- Built a production MCP server for live tea e-commerce administration, exposing 46 tools across orders, products, customers, inventory, invoices, logistics, churn analytics, and reporting with automatic token refresh and parallel API aggregation.
- Integrated Claude Desktop, Claude Code, and custom MCP clients over STDIO transport with secure environment-based credentials, structured stderr logging, and automatic recovery from expired JWT sessions.

### Qualipilot Data Quality Checker | Python, Polars, Dask, cuDF, AWS Bedrock

GitHub

- Built a production data-quality CLI/library with missing, duplicate, type, range, outlier, cardinality, and freshness checks, CI severity gates, HTML/JSON/Markdown reports, and optional LLM narration through Bedrock, Ollama, or OpenAI-compatible endpoints.
- Implemented swappable Polars, Pandas, Dask, and cuDF engines with typed Pydantic results, deterministic JSON output, Docker/Lambda deployment paths, and probabilistic record linkage for deduplication.

## EDUCATION

### Bachelor of Technology: Computer Science and Engineering

May 2023

Specialization in Artificial Intelligence and Machine Learning

GPA: 9.08

SRM Institute of Science and Technology, Chennai

## LANGUAGES

English (Fluent) | Tamil (Native) | Telugu (Fluent) | Hindi (Intermediate)